

A MODEL OF OPTIMAL SCIENCE FUNDING: TARGETING THE CAPABILITY-RESOURCE GAP

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ABSTRACT. The grant funder’s challenge is to maximize the impact of a fixed budget under uncertainty about the researchers who might receive it. We develop a Bayesian model of this challenge: researchers differ in their capabilities and in their resources; research output requires both, and funding compounds capability across rounds. We show that the optimal allocation funds the capability-resource gap: grants increase with capability and decrease with existing resources. Track records underdetermine the gap; peer review supplies an additional, noisy signal of capability. We find that this signal’s value increases with capability inequality and with the signal’s informativeness, and that under heavy-tailed capability distributions even a noisy signal from peer review captures most of that value. We further find that uniform seed grants and lotteries, which forgo targeting, lose little output exactly where review is worth little, and that whom to fund matters far more than when to fund.

Significance. Research funders divide fixed budgets among researchers whose abilities and needs they cannot observe. Whether to concentrate funds among a few researchers or spread them across many, whether peer review is worth its cost, and whether lotteries should replace it are contested, and the evidence has not settled them. We model funding as a sequential decision under uncertainty and show that the optimal allocation funds the gap between each researcher’s capability and their resources, that track records alone cannot reveal that gap, and that what peer review is worth, how lotteries and seed grants perform, and whether to concentrate or spread turn on two features of a field: the funder’s budget relative to researchers’ resources, and how unequal capability is.

The United States spent close to a trillion dollars on research and development in 2024, roughly a fifth of it funded by the federal government (National Center for Science and Engineering Statistics 2026; National Science Board 2026), and how best to allocate that money remains unsettled. Whether funding should be concentrated among a few researchers or spread across many is a long-running

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dispute (Aagaard et al. 2020). How well peer review predicts research outcomes is contested: among funded NIH grants, percentile scores predicted subsequent productivity barely better than chance (Fang et al. 2016), while studies with other designs find modest but real predictive power (Li and Agha 2015; Park et al. 2015). And several funders now allocate part of their budgets by lottery (Liu et al. 2020; Heyard et al. 2022).

These disputes concern one problem: what a funder with a fixed budget and imperfect knowledge of researchers should do. The empirical studies each measure one input to that decision, how well scores predict outcomes among funded grants or how output responds to grant size, and none can say what the funder should do with what it knows; that depends on what the funder is uncertain about, what its information is worth, and how the answer changes with the budget. We treat the funder’s problem as a sequential decision under uncertainty and analyze it with standard Bayesian methods. Researchers differ in capability, what they know and can do, and in resources, what they have to work with; output requires both and is limited by the scarcer of the two; the funder observes output, and the noisy judgments of reviewers, but neither attribute directly. A grant adds resources in the round it is given and, because doing research builds expertise, raises capability in later rounds. Each round, the funder observes the output that results, updates its beliefs about the researchers, and allocates again.

Three results follow. First, the optimal allocation funds the gap between each researcher’s capability and their resources: grants increase with capability and decrease with existing resources, and the budget sets how many researchers are funded at all. Two heuristics a funder might use, funding the track record and funding the under-resourced, each track one of the two quantities and can each produce less output than dividing the budget equally. Second, track records underdetermine the gap, and peer review supplies what they cannot, a signal of capability. The value of that signal, how seed grants and lotteries perform relative to targeted funding, and whether funds should be concentrated among a few researchers or spread across many all depend on the same two features of a field: how much of researchers’ resources the funder’s budget can supply, and how unequal capability is. Where the budget is tight and capability heavy-tailed, targeting is worth a great deal, even noisy review captures most of that value, and lotteries give up much of it; where the budget is ample and capability evenly spread, little output is at stake and spreading funds widely loses almost nothing. No funding mechanism is good or bad in itself. Third, when to fund matters far less than whom to fund: the gain from choosing the schedule of spending is smaller than the gain from a review signal in every setting we examine.

Two assumptions bound these results: grants are used up in the round they are given, with whatever lasts from them entering as capability, and capability and resources are complements. The model is deliberately simple. Its contribution is to the theoretical understanding of the funder's problem: it shows that the three disputes turn on two features of a field that a funder can estimate, and it says which way each dispute goes in each kind of field.

RESULTS

The optimal allocation funds the capability-resource gap. Consider first a funder that knows every researcher's capability K_i and resources R_i and divides one round's budget B so as to maximize expected output, which for researcher i is $2AK_iR_i/(K_i + R_i)$ with R_i the sum of baseline resources R_{i0} and the grant g_i (Materials and Methods). The marginal value of a dollar to researcher i is $2AK_i^2/(K_i + R_{i0} + g_i)^2$: it increases with their capability and decreases with the resources they already hold, grants included. An allocation is optimal exactly when no dollar can be moved to a researcher who would produce more with it, and solving that condition gives the gap rule (SI Text 3, Proposition 1):

$$g_i^* = \max(cK_i - R_{i0}, 0),$$

where $c > 0$ is the value at which the grants sum to the budget. Each researcher's target is c times their capability; the rule fills each positive gap between target and resources and grants nothing to the rest. The condition $R_{i0} < cK_i$ draws a frontier through the population, separating the funded from the unfunded (Fig. 1). When the budget is small, c is small and few researchers are funded; when the budget is large, funding extends across nearly the whole population.

The gap rule explains why each heuristic fails. Funding the track record fails because output reflects resources as well as capability: a productive, well-resourced researcher has a small or negative gap, and an additional dollar adds little to their output. Funding the under-resourced fails for the opposite reason: scarcity is no evidence of capability, and dollars given to researchers of low capability add little output however scarce their resources. Each heuristic can produce less output than dividing the budget equally (SI Text 3, Examples 1 and 2).

How much targeting matters depends on the budget. Expected output increases with resources at a diminishing rate and never exceeds $2AK_i$, so when the budget is large enough to bring every researcher's output near that bound, the optimal allocation and equal division differ little. The value of targeting, the expected output the optimal allocation adds over equal division, decreases in the budget and vanishes as the budget grows (SI Text 3, Corollary 2; Fig. S1). Targeting

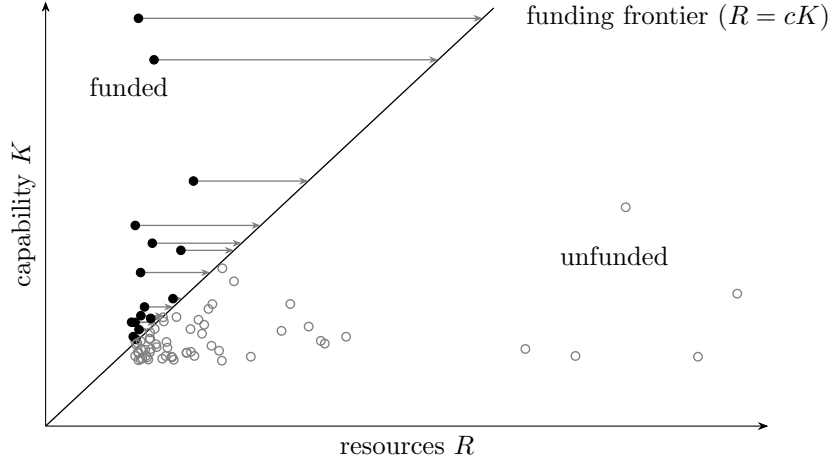


FIGURE 1. The optimal allocation funds the capability-resource gap. Each point is a researcher, placed by resources and capability; the line is the funding frontier $R = cK$ at budget scale $b = 0.1$. Researchers below the frontier (filled) are funded, and each arrow is a grant, which moves its researcher exactly to the frontier. Researchers on or beyond the frontier (open) receive nothing.

matters where the budget is small relative to researchers' resources, and this dependence on the budget recurs in every result below.

What the funder can learn: track records and peer review. A real funder observes output, not capability or resources. Expected output is one number produced by the two jointly, so a researcher of high capability and modest resources and a researcher of modest capability and abundant resources can produce identical records, and these are exactly the researchers the gap rule treats most differently: the first may have the largest gap in the population, the second none. Output gives only a lower bound on capability, $K_i > \lambda_i/2A$. The problem is worst where the gap is largest. When resources are small relative to capability, expected output is approximately $2AR_i$, proportional to resources and nearly independent of capability (Fig. 2A), so small grants reveal almost nothing about who is capable, and only grants comparable to capability itself make records informative.

A Bayesian funder relying on records alone therefore allocates far from the gap rule, and further rounds of records improve its allocation only slowly: the correlation of its grants with the optimal grants rises from 0.08 in the first round to 0.41 by round twenty, still below the 0.81 that a funder with a review signal reaches in round one (Fig. 2B). The reason lies in the model's structure. Capabilities compound while resources do not, so each researcher's output approaches the level their resources sustain, and output at that level carries less and less information about capability.

Peer review supplies what records cannot: a noisy signal of capability itself. We measure the value of review as the additional expected output of a funder allocating with the signal over the same funder allocating without it, and we report it as the share of the records-only shortfall that the signal recovers, where the records-only shortfall is the difference between the records-only funder's output and the output a funder with complete information would produce. That value depends on the field (Fig. 2C). Review is worth most where capability is heavy-tailed and the budget is tight, and under heavy-tailed capability even a signal of moderate informativeness captures most of its value: at three times the default signal noise, review retains 82% of its maximum value when the capability distribution's tail parameter is $\alpha_K = 1.3$, 44% at the default distribution ($\alpha_K = 2$), and 10% where capability is spread more evenly ($\alpha_K = 3.5$). The reason is twofold. Where capability is heavy-tailed, most of the output a funder can add comes from getting resources to the few researchers of unusually high capability, and precisely those researchers are the easiest to identify: they stand far apart from everyone else, so even a noisy signal separates them. Where capability is spread more evenly, no researcher matters much more than another, so review of any informativeness adds little; and where the budget is ample, information about whom to fund has little left to add. This ordering of review's value by field holds across production functions from Cobb-Douglas to Leontief (SI Text 6). One condition qualifies it: the funder must weight review at its true informativeness. A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether, and overtrusting review loses more output than undertrusting it (Fig. S2).

When to fund. The funder holds a fixed budget for the whole horizon and may spend it on any schedule. The strongest argument for spending early concerns fields whose researchers have few resources: a researcher without resources produces nothing, so the funder observes nothing and no capability grows, and money held for later rounds produces neither evidence nor growth in the meantime. The argument fails. A funder that spends everything in the first round allocates before observing any output, and knows exactly as little as a funder that spends everything in the last round; in a field with no resources at all, the two schedules produce identical expected output. What the argument does establish is a complementarity: early spending enables observation and growth, and late spending exploits what early spending revealed. That favors putting some money early. It does not favor putting most of it early.

Our simulations say which should come first: observation, then money. Wherever capabilities compound at more than a negligible rate (ϵ above about 0.03), the optimal schedule spends less than an even share in the first round and spends the

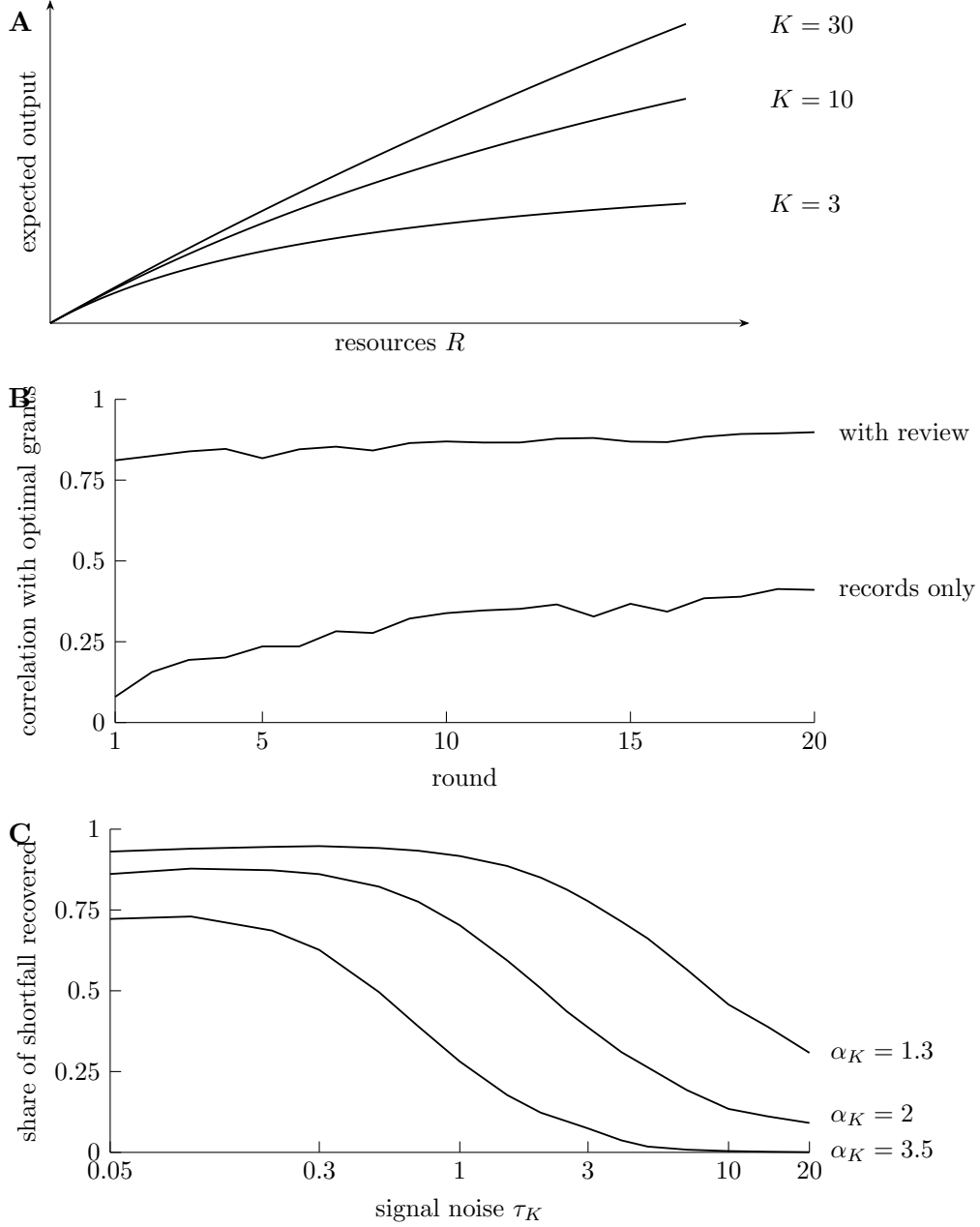


FIGURE 2. What the funder can learn. (A) Small grants are uninformative about capability: expected output against resources for researchers of capability $K = 3, 10$, and 30 ; the curves coincide where resources are small relative to capability. (B) Records do not catch up: the correlation of each funder's grants with the optimal grants, round by round; the records-only funder improves slowly and after twenty rounds remains further from the optimal allocation than the funder with the review signal is after one. (C) Review is worth most where capability is heavy-tailed: the share of the records-only shortfall that the review signal recovers, as review noise τ_K increases, for three capability distributions. Settings: SI Text 7.

bulk once output has revealed who is capable; scarce baseline resources raise the early share and weakens the effect, but does not reverse it (Fig. 3A). The shift of spending toward later rounds comes from the output researchers produce whether or not they are funded, which compounds and reveals capability while the funder waits (SI Text 5).

The choice of schedule is worth less than the signal provided by grant peer review, in every setting we test. Across all horizons and compounding rates in our sweeps, the gain from choosing the schedule, over spending equal installments each round, stays below the review signal's value: by an order of magnitude or more at the default compounding rate and below, and approaching the signal's value only at the highest compounding rate and the longest horizon we consider (Fig. 3B). As a fraction of no-funding output, the scheduling gain ranges from indistinguishable from zero to 0.7% across the 32 settings tested; the review signal's value in the same settings ranges from 0.8% to 7.5%. The gain exists only where capability and resources are complements: under Cobb-Douglas production it is indistinguishable from zero (SI Text 6). Choosing whom to fund outweighs choosing when.

Spreading funds: seed grants, lotteries, and concentration. Two alternatives to targeting recur in funding policy, and each has a rationale. Uniform seed grants give every researcher the same amount, so that a researcher whom review misjudges still receives something; in practice a funder gives out part of its budget as seed grants and targets the rest. Lotteries give grants to applicants chosen at random, usually from among those who pass a review threshold; their advocates argue that review cannot reliably rank the applicants it deems fundable, that review is costly, and that review is biased (Fang and Casadevall 2016; Roumbanis 2019). Both alternatives give up part of targeting on purpose, so both give up part of the value of targeting, and that value depends on the field. For a funder allocating with records and review, targeting adds more output than uniform funding's entire gain over no funding in a heavy-tailed field with reliable review and a tight budget, and about a quarter of that gain in the default field at the largest budget we sweep (SI Text 5). What the alternatives save, in the cost of review and in the effort researchers spend on proposals, lies outside our model (Gross and Bergstrom 2019).

Consider seed grants first. The output lost increases faster than the fraction of the budget given out as seed grants: giving out half of the budget as seed grants loses more than twice the output that giving out a quarter loses (Fig. 4A). The size of the loss is set by what targeting is worth in the field: when a review-informed funder gives out three quarters of its budget as seed grants, expected output falls

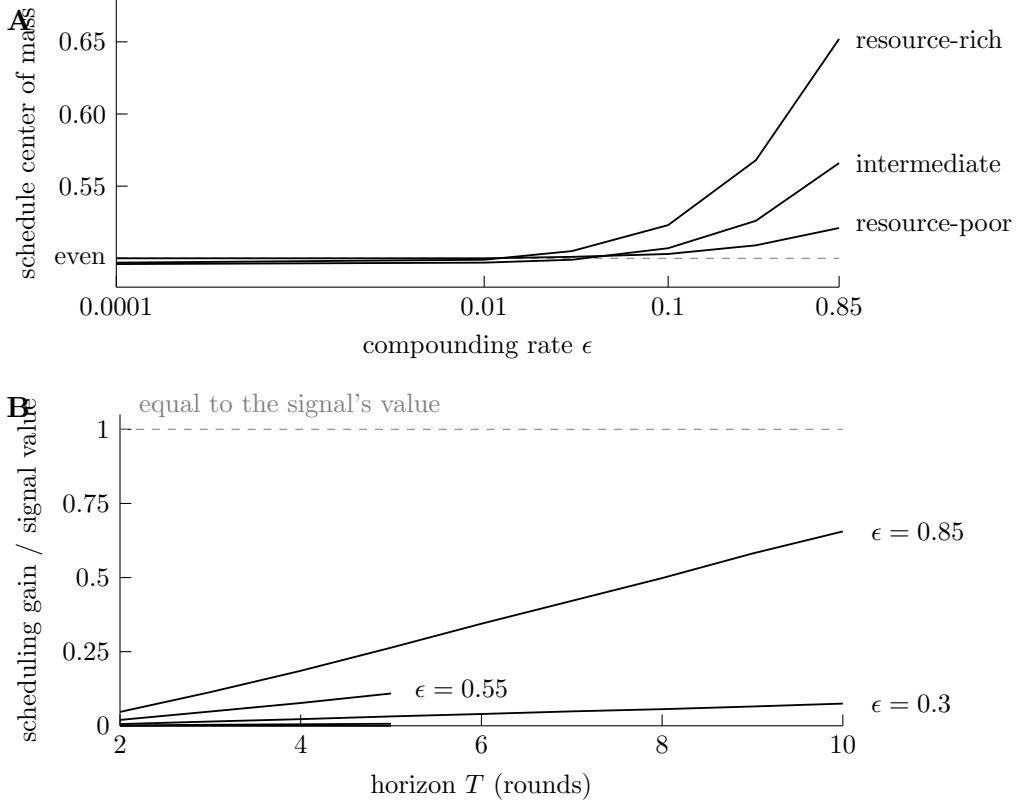


FIGURE 3. When to fund. (A) The compounding rate, not the scarcity of baseline resources, sets how far spending shifts toward later rounds: the schedule's center of mass (0.5 is even; higher is later) as the compounding rate ϵ increases, for three baseline resource scales. (B) Choosing whom outweighs choosing when: the gain from choosing the schedule, as a share of the review signal's value, by horizon and compounding rate; in every setting the gain is below the signal's value. Settings: SI Text 7.

by about 6% of no-funding output in a heavy-tailed field with reliable review, and by about 1% in the default field.

Now consider lotteries. A lottery gives each applicant in its pool a chance at a full grant instead of a certain smaller one, and on average the applicant receives the same amount either way. Because expected output increases with resources at a diminishing rate, the certain grant produces more expected output. Dividing a pool's money equally among the pool therefore produces more expected output than any lottery over the same pool, and a lottery over all applicants produces less than uniform funding (SI Text 4, Proposition 2). To measure how much less, and how lotteries compare with selection by review, we evaluate five schemes in a single round (Fig. 4B): equal division among the fifth of researchers with the highest

review scores (selection by review); equal division among the top two fifths; the partial lottery that some funders run, which funds half of the top two fifths at random with full-size grants; uniform funding; and a lottery over all researchers with full-size grants. In a heavy-tailed field with a tight budget, selection by review gains about three quarters of what the optimal allocation gains when review is reliable and about half when review is very noisy. The partial lottery gains less, and most of the difference comes from ignoring the review ranking within the top two fifths rather than from the randomness itself: at the default noise, the partial lottery falls 0.18 short of selection by review, of which ignoring the ranking accounts for 0.14 and the randomness for 0.04. The lottery over all researchers gains least, less than uniform funding. In an evenly spread field with a large budget, the ordering reverses: uniform funding gains more than four fifths of what the optimal allocation gains, and every scheme that concentrates grants, whether by review score or by chance, gains less. A lottery among applicants above a review threshold therefore loses little output only where the review ranking it ignores is worth little: replacing that ranking by chance loses a few hundredths of the optimal allocation’s gain in an evenly spread field or when review is very noisy, and about a sixth of that gain in a heavy-tailed field with reliable review. In an evenly spread field, the better alternative to review is small grants to many researchers; a lottery among a few applicants above a review threshold produces less.

Seed grants and lotteries both spread funds more widely than targeting does, and both raise the older question of whether funding should be concentrated among a few researchers or spread across many. The production function does not settle that question. Stronger complementarity between capability and resources concentrates the review-informed funder’s grants when the budget is tight in a heavy-tailed field, spreads them when the budget is ample, and has a non-monotone effect where capability is evenly spread (Fig. S4; SI Text 5). How widely funds should be spread is set jointly by the production function, the budget, and the field’s capability distribution.

DISCUSSION

Our results isolate one quantity as key to optimal grant funding: the value of targeting researchers with the greatest gap between their capabilities and their resources. Two features of a field determine this value. The first is the funder’s budget relative to researchers’ own resources: where the budget is small, dollars are scarce and where they go matters; where the budget can relieve most researchers’ bottlenecks, the optimal allocation and equal division produce nearly the same output. The second is how unequally capability is distributed: where a

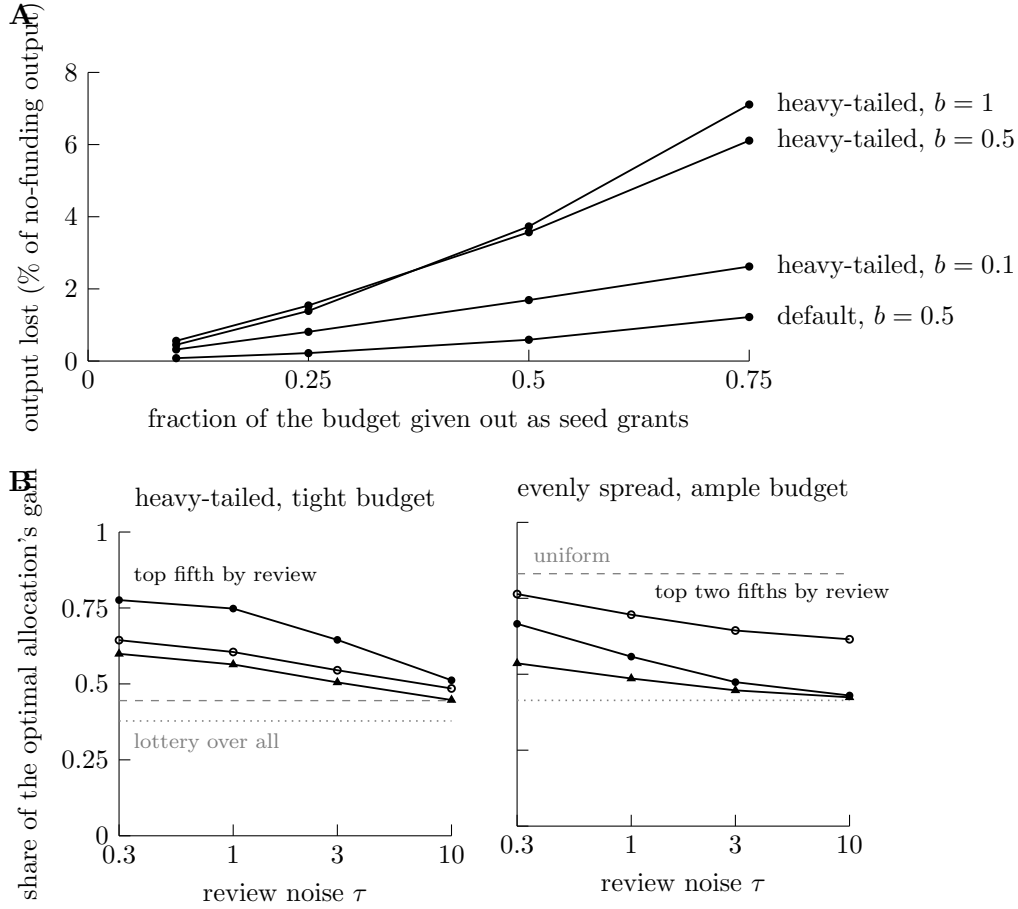


FIGURE 4. Spreading funds. (A) The output lost to uniform seed grants, as a percent of no-funding output, when a review-informed funder gives out a fraction of its budget as equal grants to all researchers and targets the rest; the loss grows faster than the fraction, and its size is set by what targeting is worth in the field. (B) What five funding schemes gain, by field: each scheme's gain over no funding as a share of the optimal allocation's gain, as review noise increases, in a heavy-tailed field with a tight budget (left) and an evenly spread field with a large budget (right). Filled circles: equal division among the top fifth by review score; open circles: among the top two fifths; triangles: the partial lottery; dashed: uniform funding; dotted: a lottery over all researchers. Settings: SI Text 7.

few researchers can produce far more than the rest, getting resources to those few is most of what a funder can add. Every result above is a statement about this quantity: what records and review contribute to capturing it, what seed grants and lotteries give up of it, and how little the timing of spending adds to it.

These results have a practical consequence for funders. No funding mechanism is good or bad in itself. Peer review, seed grants, and lotteries each perform better in some fields than in others, and the two features of the field decide which. A funder choosing among mechanisms should therefore begin by estimating the two features for its own field. The funder can compute the first, its budget relative to researchers' total resources, from its own budget and what it knows about the field. The inequality of capability is harder to observe: the natural proxy is the distribution of output, but where grants are small, output tracks resources rather than capability, so the output distribution understates capability inequality. A small foundation funding a heavy-tailed field faces the largest value of targeting per dollar. For this foundation, review directed at identifying the few researchers of unusually high capability is worth most, and even noisy review recovers most of that value; seed grants give up the most output; and a lottery among applicants above a review threshold gives up about a sixth of what selection by review gains. A large agency funding an evenly spread field faces a small value of targeting; for this agency, review of any informativeness adds little, seed grants and lotteries give up little, and the best use of the budget is small grants to many researchers.

Three further implications concern peer review. Since most of review's value comes from separating the few researchers of unusually high capability from the rest, and fine distinctions among the many applications of comparable merit add little output, a review system that spends most of its effort ranking the middle of the distribution spends that effort where the model locates the least output. Second, the finding that percentile scores barely predict productivity comes from funded NIH grants with scores in the top fifth (Fang et al. 2016), and the studies that find modest predictive power also use funded grants (Li and Agha 2015); both measure prediction among applicants that review has already selected, within the range where the model expects review to add least, so a weak association there is consistent with review of considerable value over the whole applicant pool. Third, a funder that treats review as more reliable than it is loses more output than a funder that treats review as noisier than it is, so a funder uncertain about review's accuracy should err toward undertrust.

The model holds several things fixed, and its results hold within those limits. There is one funder, and researchers do not respond strategically, so competition among funders, proposal effort, risk choice, and the cost of review are outside the model; the contest models of grant competition cover that ground (Gross and Bergstrom 2019, 2025). Output is a single number per round, so projects do not differ in kind and exploring new directions has no value in the model (Avin 2019). The review signal is noisy but unbiased; bias and conservatism in review, which are among

the strongest arguments for lotteries, are not modeled. Grants are consumed and inputs are complements; the timing results depend on complementarity and the information results do not. The numbers we report are comparisons across the parameter values we sweep, and none is calibrated to a real funder; what applies to a real field is the qualitative features of the dynamics that organize the results.

The model makes three predictions that data on funded research can test. Measures that stand in for capability, such as prior record or review score, should predict output better the larger a grant is relative to the recipient’s other resources, because on small grants output is nearly independent of capability. How well review scores predict output, measured over the whole applicant pool rather than among funded grants, should be higher in fields whose output distributions have heavier tails. And where funders run lotteries beside review, the output lost to the lottery should be larger in heavier-tailed fields. Each prediction can be tested with data that funders already hold: the first by comparing how well review scores predict output across grant sizes within one applicant pool; the second by comparing that predictive power across fields, scored over the whole applicant pool where the outcomes of funded and unfunded applicants can both be observed; the third by comparing the outputs of lottery-funded and review-funded grants where a funder runs both.

A funder with a fixed budget and imperfect knowledge of researchers faces one problem, and our model gives it one answer. The optimal allocation funds the gap between each researcher’s capability and their resources. Records alone underdetermine that gap; peer review can supply the missing signal of capability; and what review is worth, how seed grants and lotteries perform relative to targeted funding, and whether to concentrate or spread all turn on the same two features of a field. In every field we examine, choosing whom to fund matters more than choosing when. A funder deciding among review, lotteries, and seed grants should therefore begin by asking which kind of field it is funding.

MATERIALS AND METHODS

Model. A funder faces n researchers over T funding rounds. Researcher i has capability K_i and resources R_i ; initial values are drawn from Pareto distributions with tail parameters α_K and α_R (smaller values give more unequal fields) and correlation ρ . In each round, researcher i produces output drawn from a Poisson distribution with mean $\lambda_i = 2AK_iR_i/(K_i + R_i)$, the harmonic mean of the two inputs scaled by a productivity constant A ; output increases in both inputs and is limited by the scarcer. Between rounds capability compounds, $K_i \leftarrow K_i + \epsilon \lambda_i/A$, at rate ϵ . Resources do not accumulate: $R_i = R_{i0} + g_i$, where the baseline R_{i0}

recurs every round and the grant g_i is consumed in the round it is given. The funder’s total budget over the horizon is $2bn\mathbb{E}[R_0]$, so at budget scale $b = 0.5$ it equals the community’s expected baseline resources for one round; the budget does not depend on the horizon.

Information and strategies. The funder knows the distributions but observes no researcher’s capability or resources. It observes each researcher’s output as it accumulates; a resource signal, observed once, gives a noisy estimate of baseline resources (noise τ_R); a review signal, drawn afresh each round it is used, gives a noisy estimate of capability (noise τ_K). Bayesian strategies update beliefs about every researcher after each round and allocate each dollar to the researcher whose expected output, given current beliefs, rises most with it. They differ in information (with or without the review signal) and in planning (myopic, optimizing the current round, or forward-looking, re-planning the remaining budget over the remaining horizon each round). Three baselines complete the set: no funding, against which every result is normalized; uniform funding; and track-record funding, which allocates in proportion to the previous round’s output. Seed grants and lotteries modify these strategies as described in Results (SI Text 2, Table S1).

Parameters and scoring. Tail parameters range from 1.3 to 3.5, covering the Pareto tails near 1 to 2 estimated for research productivity (Lotka 1926; Nicholls 1989) and extending to much more equal fields; the budget scale b from 0.1 to 1, with larger budgets in the timing analyses; the compounding rate ϵ from near zero to 0.85; review noise τ_K from 0.05 to 20; the horizon from one round to ten. Default values are $T = 2$, $n = 50$, $\epsilon = 0.1$, $b = 0.5$, $\alpha_K = \alpha_R = 2$, $\tau_K = \tau_R = 1$, $\rho = 0$. Each strategy is scored by the sum of expected outputs along its run, which equals realized output averaged over repetitions of the same allocations, and strategies are compared on identical simulated populations with identical signal and output draws, so comparisons are paired. Results are reported as each strategy’s improvement over no funding, in percent, or as shares of a named comparison. Analytic results (the gap rule, its corollaries, and the lottery proposition) are proved in SI Text 3 and 4; the settings behind each figure are in SI Text 7. Code and data are available in the project repository.

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